TROT DYNASTY · RACE AND BREEDING ENGINES · REVISED WITH SIMULATION

ENGINE REVIEW

A review of Astra's outline for the race and breeding engines against the code running trotdynasty.com. The live world has too few players to judge, so every finding here comes from simulation. The current world was run forward 30 seasons per variant, twice each, about 50,000 races per run. Separately, 20,000 test races measured the race engine itself. PC copy only; nothing touched the live server.

Verdict, revised: Astra's outline holds up, but the simulations moved the priorities.

- **Ratings inflate, and genetics isn't the cause.** Foal potential stays flat for 30 seasons. Yet the share of racers graded S or S+ climbs from 19% to 25% with only AI barns breeding. The creep comes from how horses develop with training and age.
- **Breeding is too random to reward skill.** Parents barely predict their foals, and strategic breeding barns finished no better than the AI.
- **One mating habit causes the sire monopoly.** Five stallions sire about three quarters of all racers in every season of every run.
- **The race engine is fair but flat.** Ability counts, but running style never matters, and in evenly matched fields the result is close to a lottery.

WHAT THE TEST LEAGUE DID

Each run started from the live world and ran 30 seasons of full race nights under four set-ups:

VARIANT	WHAT IT ADDS
Baseline	Today's rules, AI barns only.
Strategic	12 barns with $5M extra, run like serious players. Three each always use the top sire, breed for speed, cover the mare's weakness, or breed close.
Fixes, first try	Strategic barns plus my first proposals: books of 30 covers, AI mares spread across the top 12 stallions, and a cost for close breeding.

Predictable	Strategic barns plus the same books and spreading, no close-breeding cost, and foals that follow their parents more closely.

SEASON 30, SIDE BY SIDE

MEASURE (AVERAGE OF 2 RUNS)	START	BASELINE	STRATEGIC	FIXES, FIRST TRY	PREDICTABLE
Average racer, true rating	48.7	53.4	48.0	35.2	54.7
Racers graded S or S+	19%	25%	17%	6%	29%
Racers graded S+	6%	12%	6%	2%	14%
Newest crop, average potential	−2.6	−2.3	−2.7	−5.2	−1.7
Racers sired by the top 5 stallions	73%	76%	75%	44%	47%
How well parents predict a foal (0 to 1)	0.12	0.12	0.14	0.03	0.10
Rating favorite wins	—	56%	55%	57%	57%

"How well parents predict a foal" is the correlation between the parents' average peak and the foal's peak. 0 means no link; real racehorse pedigrees sit around 0.3 to 0.4.

FINDINGS

+5

RATINGS CREEP WITHOUT PLAYERS

The average racer's true rating rises about five points over 30 seasons in the baseline. The S and S+ share climbs from 19% to 25%. Foal potential stays flat, so the creep comes from development: training and age curves, not genes.

0.12

PEDIGREE BARELY PREDICTS A FOAL

Good parents hardly improve the odds. The four strategic breeding plans finished within a point of each other and of the AI. Breeding skill earns almost nothing today.

~75%

THE SIRE MONOPOLY IS STRUCTURAL

AI barns send 70% of their mares to the best stallion with room, and a book holds 100 covers. Five sires hold three quarters of the racing population in every season of every baseline run.

Halved

SPREADING MARES FIXES THE MONOPOLY

Books of 30 plus AI mares spread across the top 12 stallions cut the top-five share from 76% to 44–47%. Nothing else moved it.

−13

A CLOSE-BREEDING COST COLLAPSED THE HERD

Because so few sires dominate, almost every pairing shares ancestors. Even a small penalty hit every foal: average true rating fell from 48.7 to 35, and S nearly vanished. A close-breeding cost can only come after the monopoly is broken, and must be tiny.

+1 to +3

PREDICTABLE BREEDING REWARDS SKILL

With foals following their parents more closely, the strategic barns' horses improved by up to three points over the AI average. The top crop rose too, and inflation sped up only slightly.

THE RACE ENGINE ON ITS OWN

TEST	RESULT	READING
Better horse beats the worse, gap 0–2 points	52%	A coin flip, as it should be.
Same, gap 10–12 points	73%	Ability tells, steadily.

Same, gap 20+ points	95%	A class edge almost always wins.
Strongest horse wins an evenly matched 8-horse field	17%	Random would be 12.5%. Close fields are near-lotteries.
Same field rerun 40 times: share for its most frequent winner	47%	Results repeat about half the time.
Post 1 wins, half-mile track	23%	Post 8 wins 5%. Strong, as on real half-mile tracks.
Post 1 wins, mile track	16%	Post 8 wins 10%. Mild.
Front-runners: share of wins against share of starters	13% vs 14%	Every style wins in proportion to how often it starts. No niches.
Breaks per start	2.3%	About one in 43. Plausible.

The engine is fair and monotonic: more ability means more wins, and post position counts. Astra's critique still lands on what it can't do. A trip is a label, not something that happened, so running style never decides a race. In matched fields the result comes down to one noise roll.

SECTION BY SECTION, RE-ASSESSED

ASTRA'S POINT	STATUS	WHAT THE SIMULATIONS SAY
1. Capabilities, not one score	PARTLY	Five traits exist, but styles win in exact proportion to how often they start. The traits don't create niches. A step-by-step race engine is what would make them matter.
2. Simulate the trip	NO	Confirmed as the largest gap. Close races come down to the noise roll, and a horse never loses because of a move.
3. Uncertainty in specific places	PARTLY	Big gaps behave well. Small gaps are close to random. The single noise roll carries too much weight between near-equals.
4. Breed potential, not records	PARTLY	The structure is right, but pedigree predicts too little, around a third of a realistic level. Skill can't show through it.
5. Uncertainty won't stop creep	WRONG CAUSE	Creep happens, but genetics isn't driving it. Development is. Breeding fixes alone won't stop it.
6. Progress through understanding	PARTLY	With pedigree this weak, reading a pedigree isn't a skill yet. That has to change first.

7. Build the test environment	NOW BUILT	The test league exists and has already caught one harmful fix, the close-breeding cost, before it shipped.

Correcting my earlier advice. For batch 2, I recommended a stronger pull to average for foals. The data says the pull is already too strong: it is why breeding skill earns nothing. I now recommend the opposite, a slightly weaker pull, paired with a hard cap on S+ and a fix for development creep. I also withdraw the close-breeding cost until the sire monopoly is broken.

REVISED PLAN

STEP 1 · BREAK THE MONOPOLY · DAYS, LOW RISK

SPREAD THE MARES AND SHRINK THE BOOKS

Books of 30 covers, and AI mares spread across the leading stallions instead of 70% to the leader. The only change that moved the top-five share, from 76% to under 50%. It also slows stud-fee runaway for players.

STEP 2 · STOP THE CREEP AT ITS SOURCE · DAYS

FIND AND CAP DEVELOPMENT INFLATION

Measure which part of development adds the five points: training gains, racing gains or the age curve. Cap it so the average racer holds steady across seasons. Add the hard S+ cap by count from batch 2 as a backstop.

STEP 3 · MAKE BREEDING A SKILL · DAYS

PEDIGREE THAT MEANS SOMETHING

Raise how closely foals follow their parents, aiming for a parent-foal link around 0.3. In the predictable variant this rewarded strategic barns without runaway genetics. Re-test with the creep fix in, so the two don't stack.

STEP 4 · THE RACE ENGINE · WEEKS

A STEP-BY-STEP RACE, BEHIND A SWITCH

As in the first report: positions, effort and cover carried call by call, with the replay reading the record directly. New targets from the probe:

- Running styles should win more where the race shape suits them.
- An evenly matched favorite should win more like 25–30%, not 17%.
- Post effects and the big-gap win rates stay as they are.

STEP 5 · AFTER THAT

INSTRUCTIONS, RACE EVIDENCE, NICHE RACES, AND CLOSE-BREEDING COST

Driver instructions, chart-caller notes, sprint and stamina races, and finally a tiny close-breeding cost once no five sires own the herd.

Every step gets re-run through the test league before it ships, with the same measures as above.

DECISIONS FOR YOU

1. **Break the monopoly first?** My pick: yes, books of 30 and spread AI mares.
2. **Reverse the batch 2 plan?** My pick: drop the stronger pull to average, keep the S+ cap, add the development-creep fix.
3. **Make pedigree more predictive?** My pick: yes, aiming for a parent-foal link around 0.3, tested with the creep fix.
4. **Race engine targets?** My pick: niches for running styles and a more decisive favorite among equals, with today's post and big-gap behavior kept.
5. **Close-breeding cost?** My pick: hold until the monopoly is gone. The first try collapsed the herd.

Limits: two random seeds per variant; strategic barns breed but don't price their stallions the way real players would; stud-fee income wasn't measured; players aren't simulated racing or buying on the market.